EXPERIMENT 8: TEXT GENERATION USING HUGGING FACE INFERENCE API

Experiment / Practical Record

1. Aim

To develop a Python program that uses a Hugging Face model through the Inference API to generate text from user-provided prompts.

2. Procedure

1. Open Google Colab and create a new notebook.
2. Install the required Python library, `huggingface_hub`.
3. Create a Hugging Face account and generate an access token.
4. Import the required library and configure the API token.
5. Select a text-generation model supported by the Inference API.
6. Enter a prompt from the user.
7. Send the prompt to the API and generate a response.
8. Display the generated text output.

3. Python Code (Google Colab — Single Cell)

```
# Experiment 8: Text Generation Using Hugging Face Inference API

!pip install -q huggingface_hub

from huggingface_hub import InferenceClient
from getpass import getpass

# Enter Hugging Face access token securely
token = getpass("Enter your Hugging Face token: ")

# Initialize the Inference API client
client = InferenceClient(
    model="Qwen/Qwen2.5-7B-Instruct",
    token=token
)

# Get prompt from user
prompt = input("Enter your prompt: ")

# Generate text
response = client.chat_completion(
    messages=[{"role": "user", "content": prompt}],
    max_tokens=200
)

# Display generated response
print("\\nGenerated Text:")
print(response.choices[0].message.content)

print("\\nResult: Text generation using Hugging Face Inference API completed.")
```

4. Google Colab Output Screenshot (Illustrative)

The image below is an illustrative Google Colab-style output preview, not a screenshot captured from a live Colab execution. Actual responses can vary, and model access depends on the current Inference API availability for your account.

Google Colab

Sample output preview

Experiment 8: Hugging Face Inference API

```
!pip install -q huggingface_hub
from huggingface_hub import InferenceClient
client = InferenceClient(model='Qwen/Qwen2.5-7B-Instruct', token=token)
prompt = input('Enter your prompt: ')
response = client.chat_completion(...)
```

Output

```
Enter your Hugging Face token: ********
Enter your prompt: Explain Artificial Intelligence.

Generated Text:
Artificial Intelligence (AI) is a branch of computer science
that enables machines to perform tasks that typically require
human intelligence, such as learning, reasoning, and
problem-solving.

Result: Text generation using Hugging Face Inference API completed.
```

Illustrative sample only — actual API output may vary.

5. Result

Thus, a Python program was developed using the Hugging Face Inference API to generate text from user-provided prompts and display the generated response.

Note

Create or manage an access token at https://huggingface.co/settings/tokens. Keep your token private. If the selected model is unavailable through the Inference API, choose another model that is supported for your account.